

Software Testing & DevOps CI/CD - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Software Engineering | Chapter: Chapter 3: Software Testing, Quality Assurance & DevOps CI/CD | Academic Year: 2026-2027

Testing Hierarchy & Best Practices

- * Unit Tests: Fast, isolated, test single class/method. Best framework: JUnit 5 + Mockito.
- * Integration Tests: Validate interactions across components/databases. Best: Testcontainers, SpringBootTest.
- * TDD (Test-Driven Development): Red -> Green -> Refactor cycle.

Black-Box vs White-Box Testing

- * Equivalence Partitioning: Group inputs where system treats all members identically.
- * Boundary Value Analysis: Bugs cluster at boundaries (e.g. testing 0, 1, 99, 100, 101).
- * Cyclomatic Complexity: $V(G) = \text{Edges} - \text{Nodes} + 2$. Indicates the number of independent execution paths.

DevOps & CI/CD Pipeline Stages

- * 1. Commit -> 2. Build -> 3. Unit Test -> 4. Static Code Analysis (SonarQube) -> 5. Package (Docker) -> 6. Deploy.
- * CI ensures bugs are caught within minutes of code submission.
- * Immutable Infrastructure: Deploying pre-built containers rather than mutating running servers.